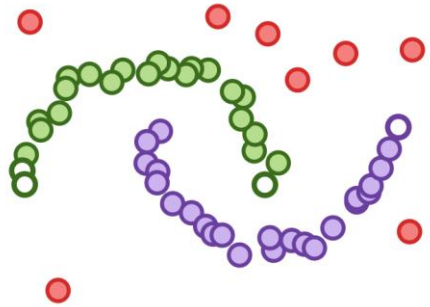
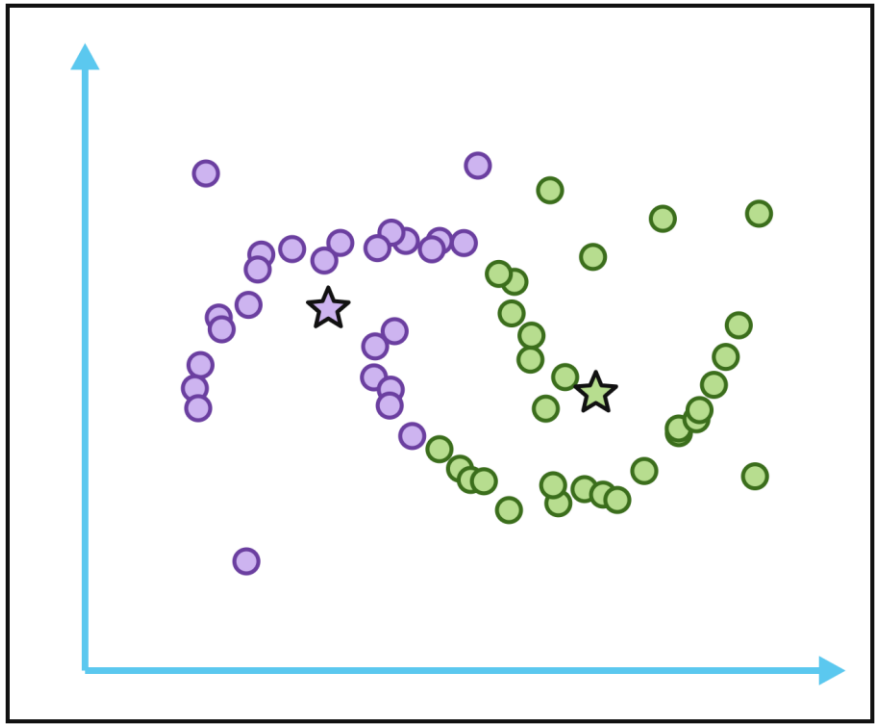
Density-Based Clustering: DBSCAN

Clusters as dense regions, separated by sparse ones



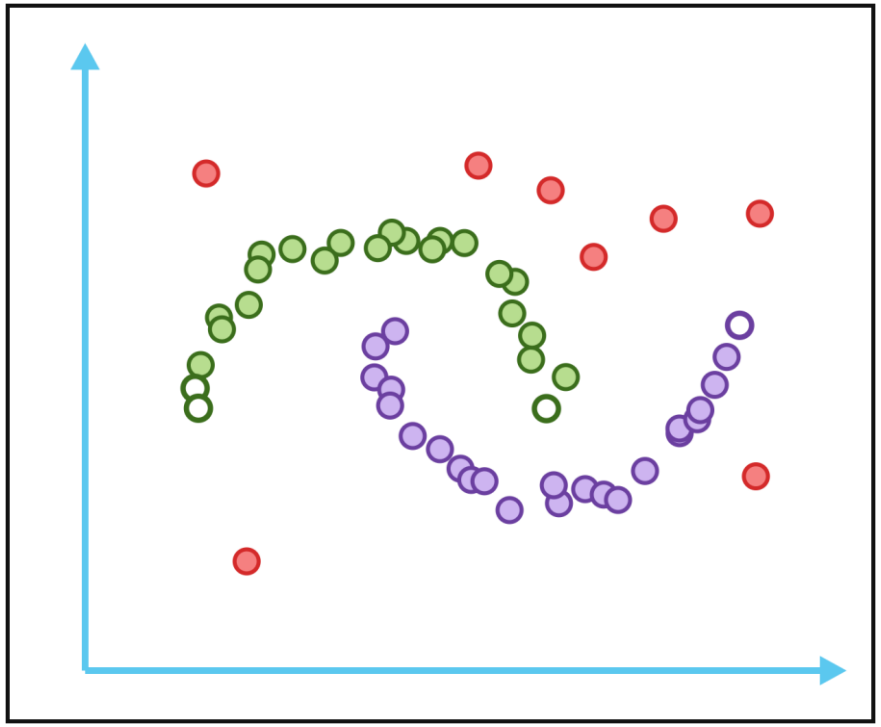
Where k-Means Struggles

GPS points from walkers on two winding trails, plus stray readings



k-means, k = 2

Cuts across both trails; strays forced into clusters.

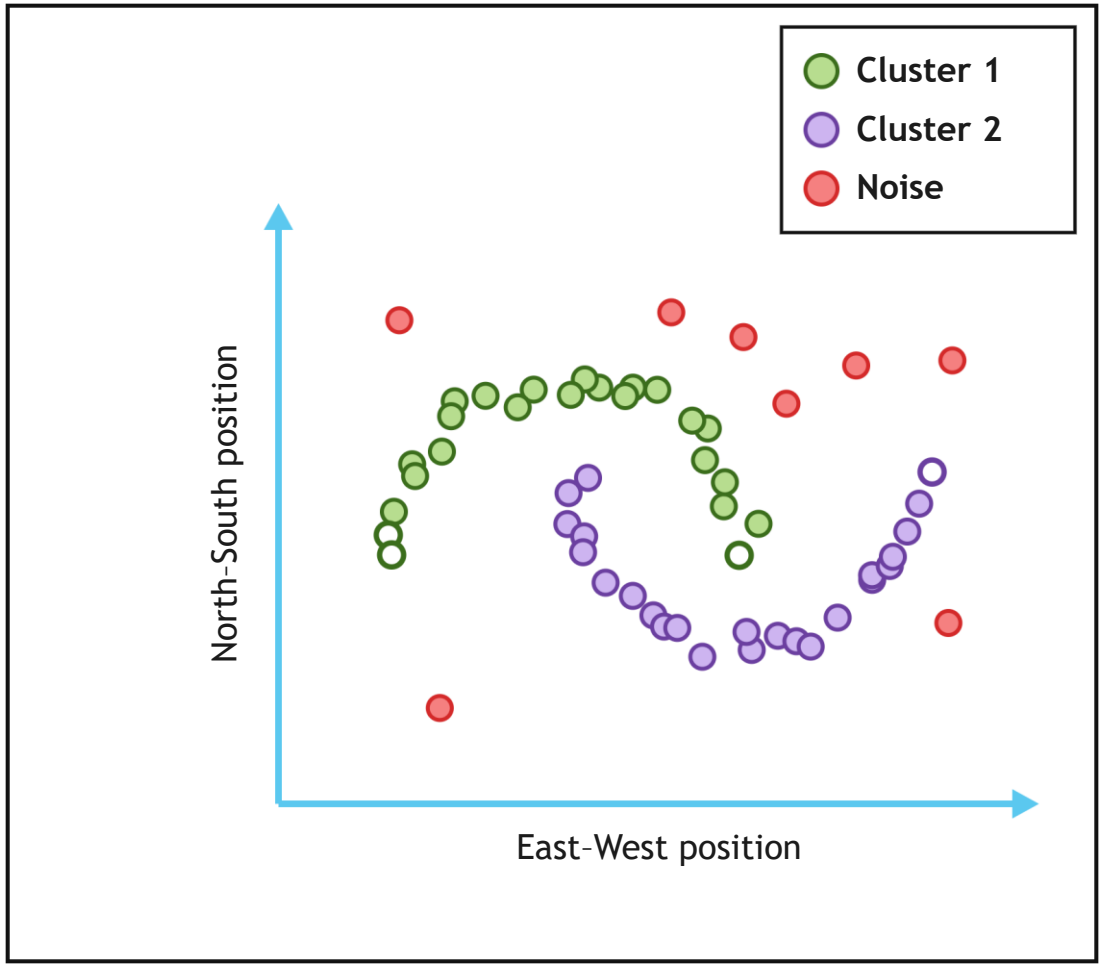


DBSCAN

Follows each trail; 8 stray points marked as noise.

What Is Density-Based Clustering?

Clusters are dense regions; sparse points are noise



Dense regions

A cluster is a region where points are packed closely together.

Any shape

Clusters follow the data, so curved or long shapes are fine.

Noise

Points in sparse areas join no cluster: they are labelled noise.

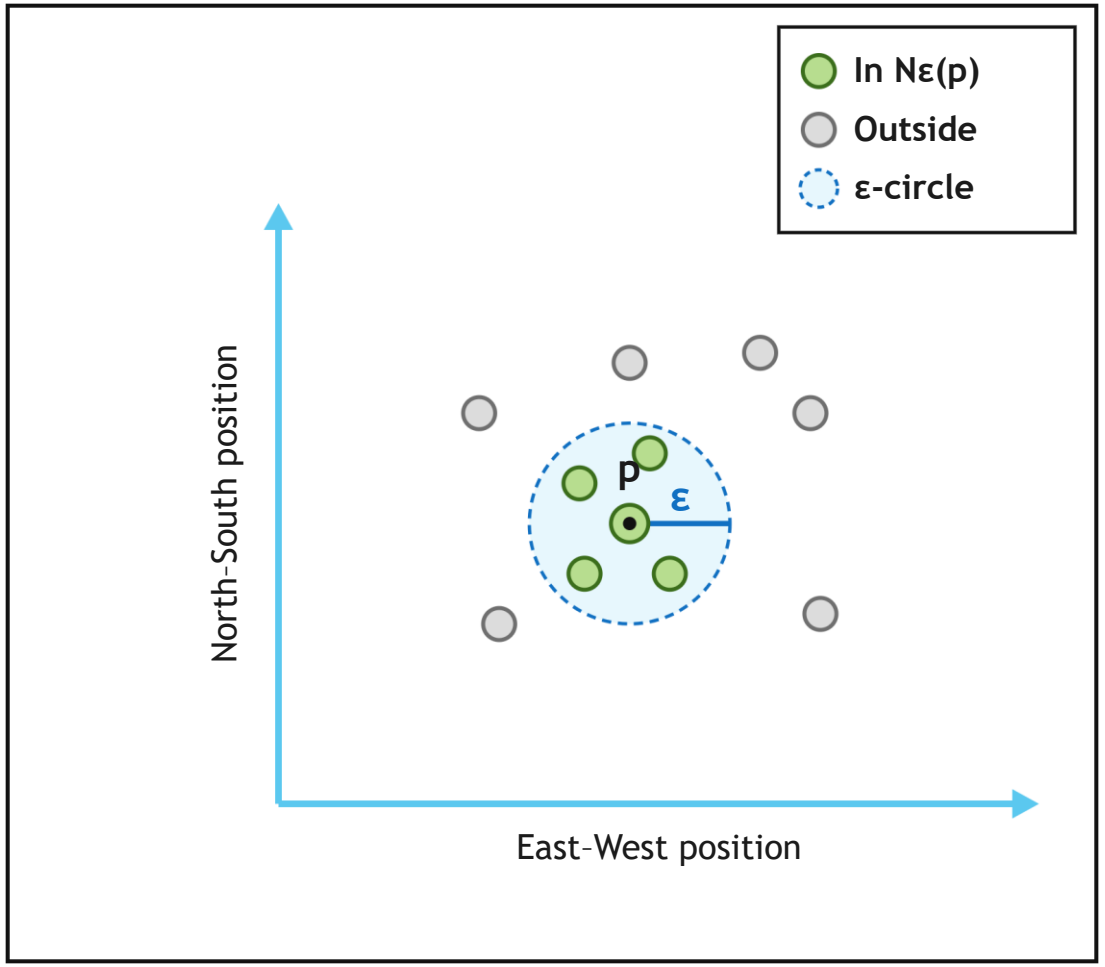
Part 1 of 3

Core Ideas

Neighbourhoods, point types and reachability

Two Parameters: ε and MinPts

A circle of radius ε, and how many points it must hold



ε (Eps)

The radius of the neighbourhood around each point.

MinPts

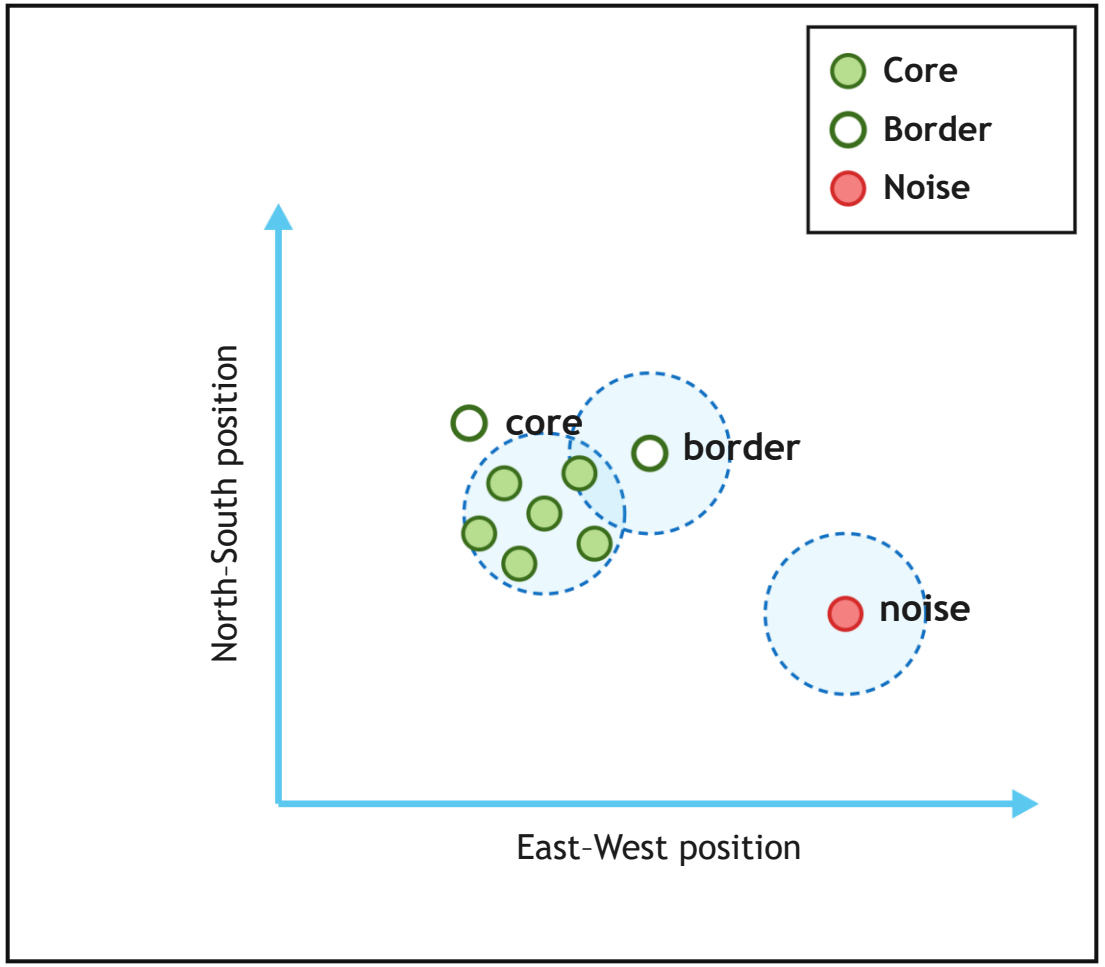
Minimum number of points, the point itself included, inside that radius.

Here

Nε(p) holds 5 points. With MinPts = 4, p sits in a dense region.

Core, Border and Noise Points

Every point gets one of three roles



Core

At least MinPts points in its ε-circle (here 6 ≥ 4).

Border

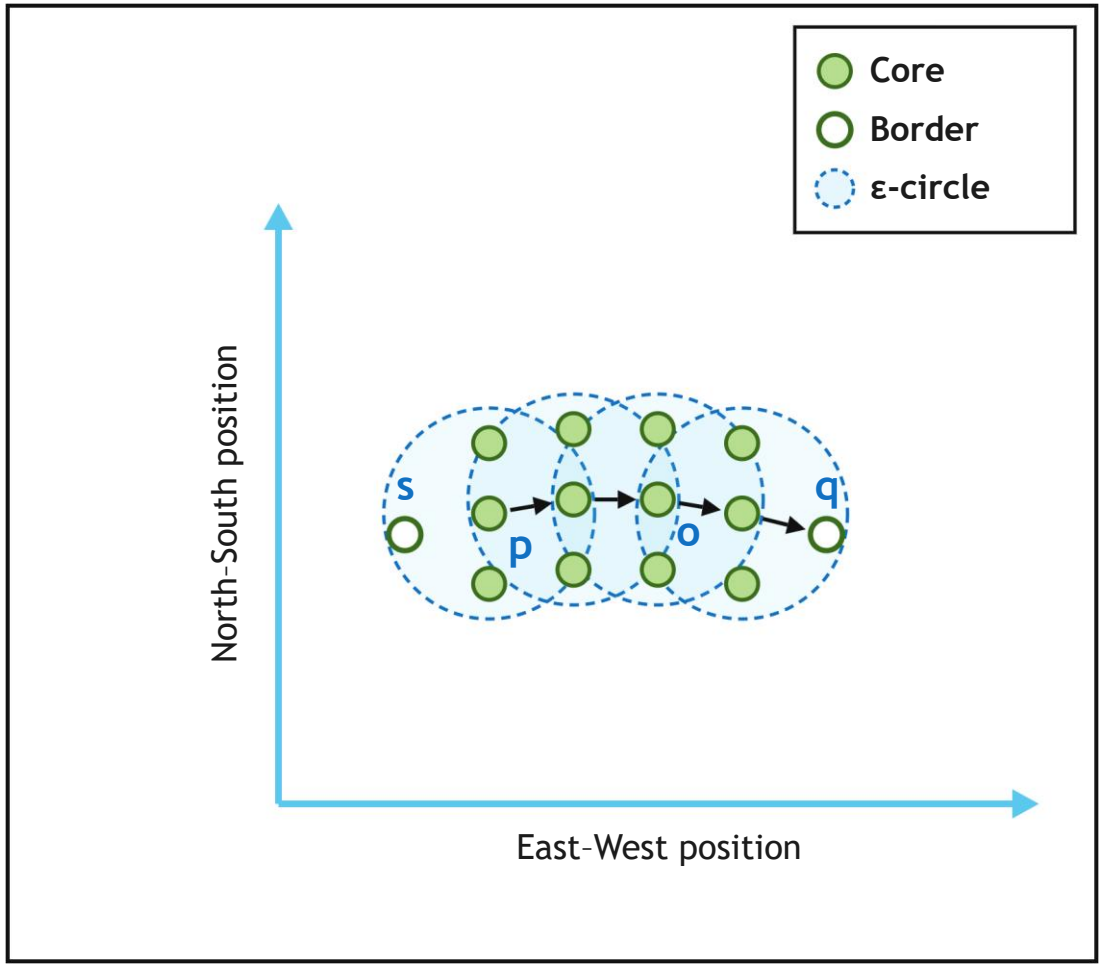
Too few neighbours to be core, but inside the ε-circle of a core point.

Noise

Neither core nor border: in a sparse region on its own.

Growing a Cluster: Reachability

Clusters spread from core point to core point



Directly reachable

q lies inside the ε-circle of a core point.

Reachable

A chain of core points leads from p to q (the arrows).

Connected

s and q are both reachable from core point o.

Cluster

A maximal set of density-connected points.

The DBSCAN Algorithm

Find a core point, then flood out through its neighbours

1	**Visit**	Pick an unvisited point p and find its ε-neighbourhood.
2	**Check**	Fewer than MinPts points? Mark p as noise for now; go to step 1.
3	**Start**	Otherwise p is core: start a new cluster with p and its neighbours.
4	**Expand**	For each neighbour that is also core, add its neighbours too.
5	**Repeat**	Continue until every point has been visited.

A point first marked noise can later be claimed as a border point of a cluster.

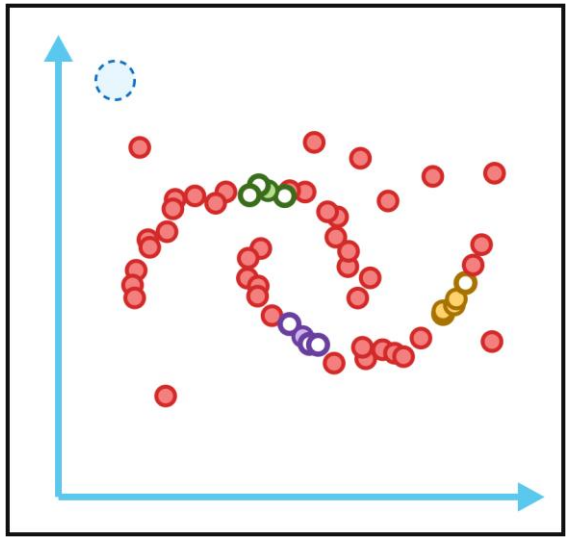
Part 3 of 3

Choosing ε and MinPts

The two settings that decide the result

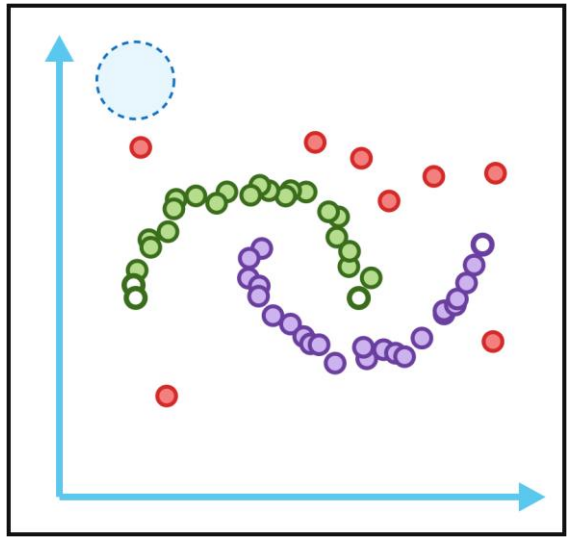
How ε Changes the Result

Same trail data, MinPts = 4, three values of ε



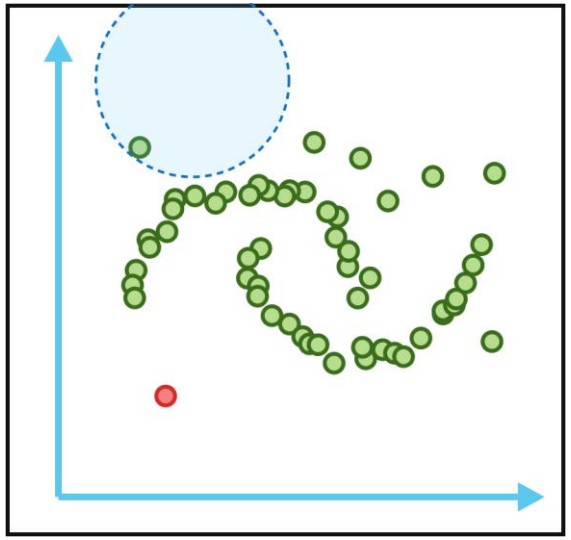
ε too small

3 clusters, 43 noise: the trails break apart.



ε about right

2 clusters, 8 noise: both trails found.



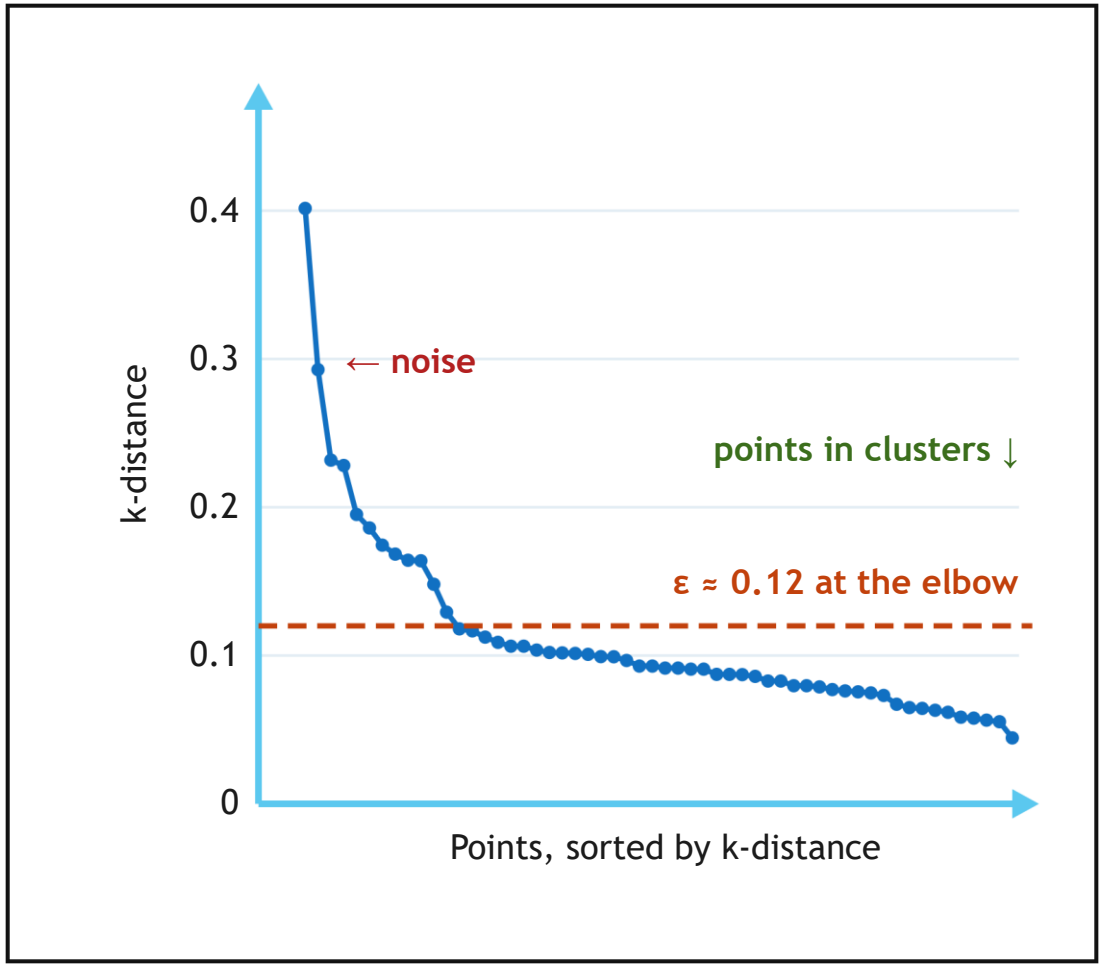
ε too large

1 cluster, 1 noise: everything merges.

Size of ε (top left) · Noise · Other colours: one per cluster · MinPts = 4 throughout

Choosing ε: The k-Distance Plot

Sort each point's k-th neighbour distance and look for the elbow



k-distance plot

Each point's distance to its k-th nearest neighbour (k = MinPts − 1 = 3), sorted high to low.

Pick ε at the elbow

Left of it: sparse noise. Right: points inside clusters.

MinPts

At least dimensions + 1; 2 × dimensions is a common start (here 4).

DBSCAN: Strengths and Weaknesses

Flexible shapes and noise handling, at the cost of tuning

Strengths

- No need to choose the number of clusters
- Finds clusters of any shape
- Detects noise and outliers explicitly
- Only two parameters; core and noise points are deterministic

Weaknesses

- Sensitive to the choice of ε and MinPts
- One ε cannot fit clusters of very different density
- Distances lose meaning in high dimensions
- $O(n^2)$ without a spatial index

Beyond DBSCAN

OPTICS and HDBSCAN handle varying density by considering many values of ε.

Three Families Compared

Partitioning, hierarchical and density-based

	k-Means	Agglomerative	DBSCAN
Needs k?	Yes	No: cut the tree	No
Cluster shape	Round, similar size	Depends on linkage	Any shape
Noise	Every point assigned	Every point assigned	Labelled as noise
Parameters	k (and start)	Linkage, cut height	ε and MinPts
Cost	O(nkt)	$O(n^2)$ memory, ≥ $O(n^2)$ time	O(n log n) with an index; $O(n^2)$ worst
Struggles with	Shapes, outliers	Large n	Varying density, high dimensions

Key Takeaways

Clusters as dense regions

1 Density

Clusters are dense regions; points in sparse regions are noise.

2 Two settings

ε (radius) and MinPts (how many points make it dense) decide everything.

3 Algorithm

Find a core point, flood out through reachable core points, repeat.

Use DBSCAN when clusters may have odd shapes, outliers matter, and k is unknown. Choose ε with a k-distance plot.